Intuitively Understanding the Shannon Entropy

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https://www.youtube.com/watch?v=0GCGaw0QOhA

- Some of the slider are from:
  "A Gentle Tutorial on Information Theory and Learning"
  by Roni Rosenfeld, Carnegie Mellon University.

$$S = k_b \ln \Omega$$

Entropy

Measures disorder

- In physical sciences entropy is used to measure the disorder of a system.
- Based on terms such as chaos and randomness.

$P(x_i)$: probability of the outcome x_i

$P(x)$: probability mass function of the distribution.

$$S = k_b \ln \Omega$$

Entropy

Measures disorder

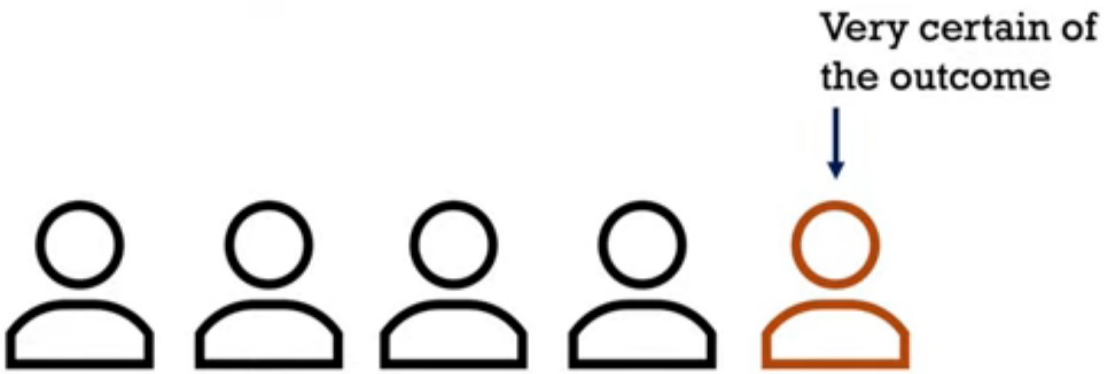
$$H(X) = -\sum_i P(x_i) \log P(x_i)$$

Shannon Entropy

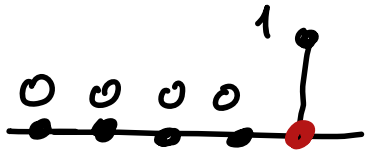
Measures the uncertainty of a probability distribution

- Discuss Shannon entropy from the point of view of information transfer.





- Probability that the next US president war born in US?

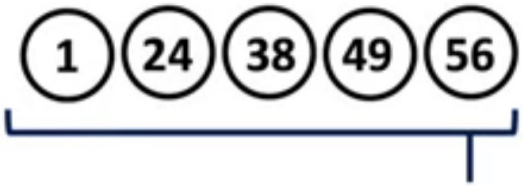


We are <u>certain</u> about the outcome.









- The distribution of winning lotto numbers

○ ○ ○ ○ ○

5 numbers btwn 0-99 $\longrightarrow$ $(100)^5$ different numbers with $\frac{1}{(100)^5}$ probability each.

uniform distribution, very uncertain

Information

- information $\neq$ knowledge
  Concerned with abstract possibilities, not their meaning
- information: reduction in uncertainty

Imagine:

#1 you're about to observe the outcome of a coin flip

#2 you're about to observe the outcome of a die roll

There is more uncertainty in #2

Next:

1. You observed outcome of #1 $\rightarrow$ uncertainty reduced to zero.
2. You observed outcome of #2 $\rightarrow$ uncertainty reduced to zero.

$\Longrightarrow$ more information was provided by the outcome in #2

Definition of Information

(After [Abramson 63])

> Let E be some event which occurs with probability $P(E)$. If we are told that E has occurred, then we say that we have received
>
> $$I(E) = \log_2 \frac{1}{P(E)}$$
>
> bits of information.

- Base of log is unimportant — will only change the units
  We'll stick with bits, and always assume base 2
- Can also think of information as amount of "surprise" in E
  (e.g. $P(E) = 1, P(E) = 0$)
- Example: result of a fair coin flip ($\log_2 2 = 1$ bit)
- Example: result of a fair die roll ($\log_2 6 \approx 2.585$ bits)

Information is Additive

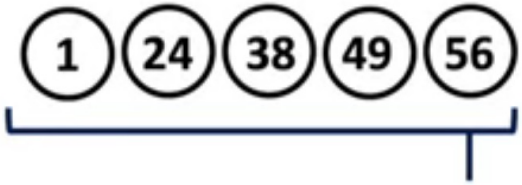
- $I(k$ fair coin tosses$) = \log \frac{1}{1/2^k} = k$ bits
- So:
  - random word from a 100,000 word vocabulary:
    I(word) $= \log 100,000 = 16.61$ bits
  - A 1000 word document from same source:
    I(document) = 16,610 bits
  - A 480x640 pixel, 16-greyscale video picture:
    I(picture) $= 307,200 \cdot \log 16 = 1,228,800$ bits
- $\Longrightarrow$ A (VGA) picture is worth (a lot more than) a 1000 words!
- (In reality, both are gross overestimates.)





little information



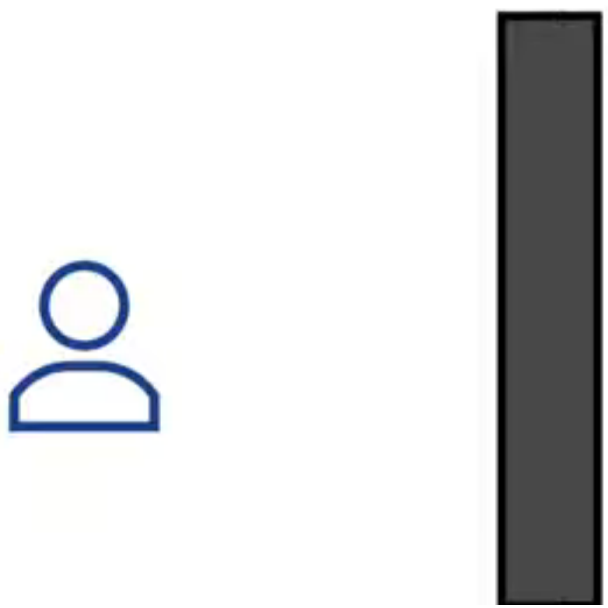


big information

Very uncertain
of the outcome

how much information, on average, would we need to encode an outcome from the distribution

- Imagine we're on one side of a wall where a random outcome occurs.



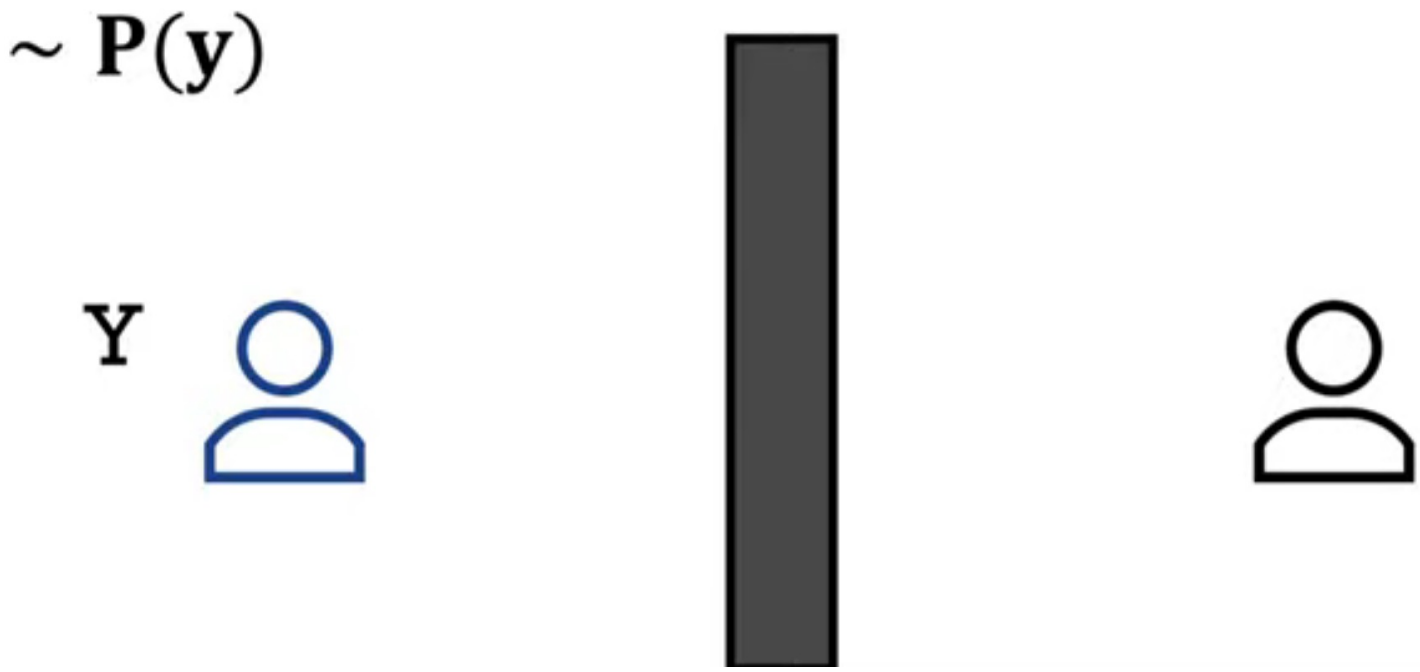
- The random outcome Y comes from the probability distribution $P(y)$.

$$Y \sim P(y)$$

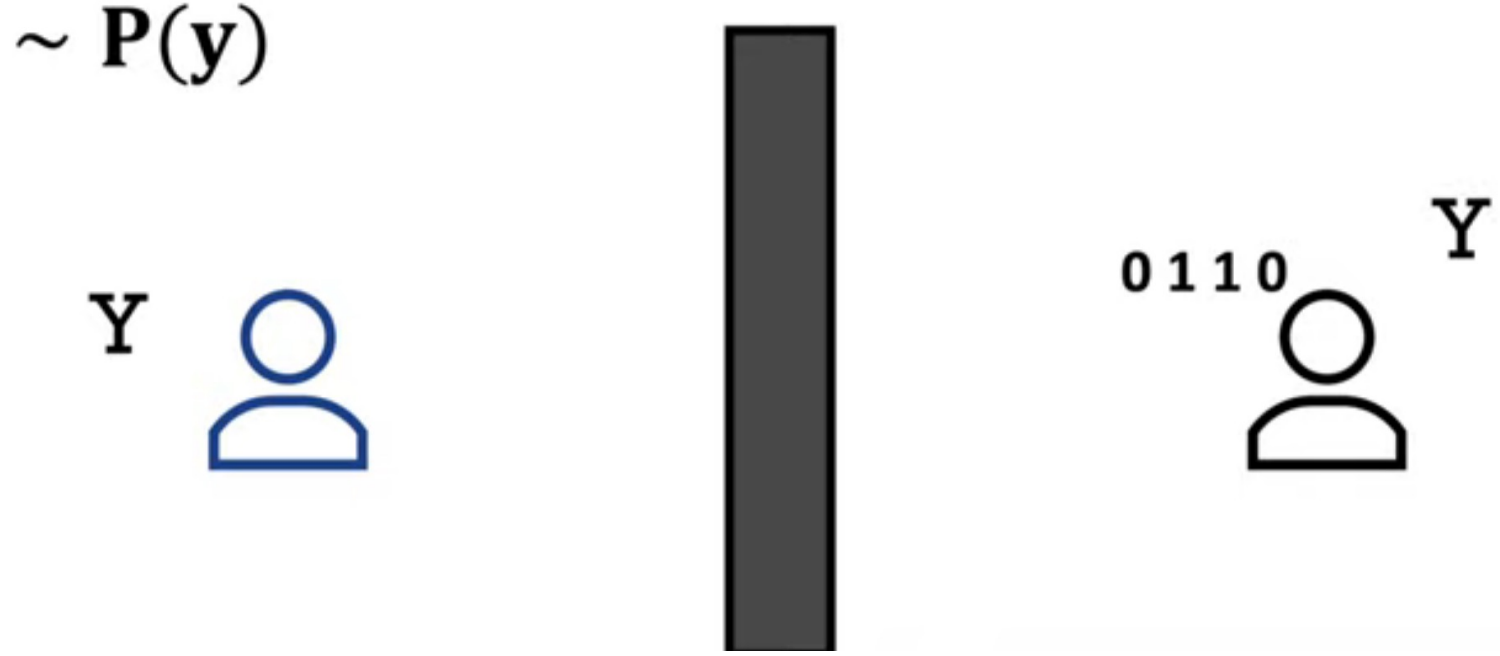




- There's another person on the other side of the wall who can not observe the outcome.



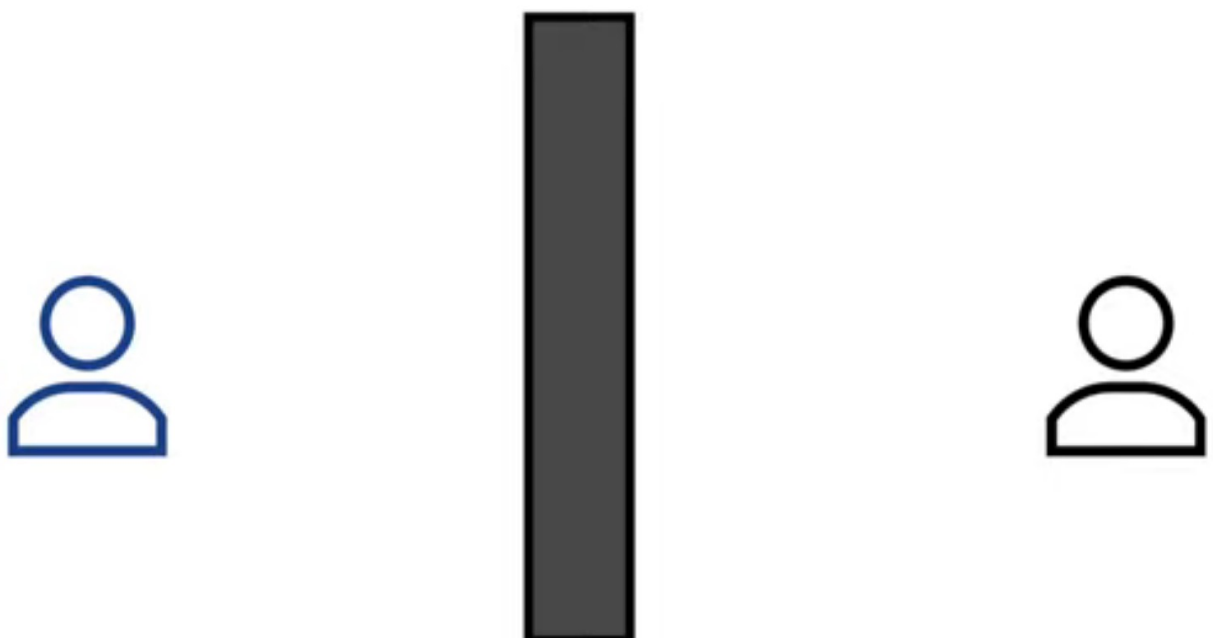
- Goal: Let the person know the outcome Y by sending an N bit number across the channel.
- Before the experiments both people agree on the mapping of n bits to a real event.
- N bit number is the information.



- Let the random event be a coin flip.

$P(H) = 0.5$

$P(T) = 0.5$



- We only need a single bit to describe the outcome:

P(H) = 0.5

P(T) = 0.5

0 = Tails

1 = Heads







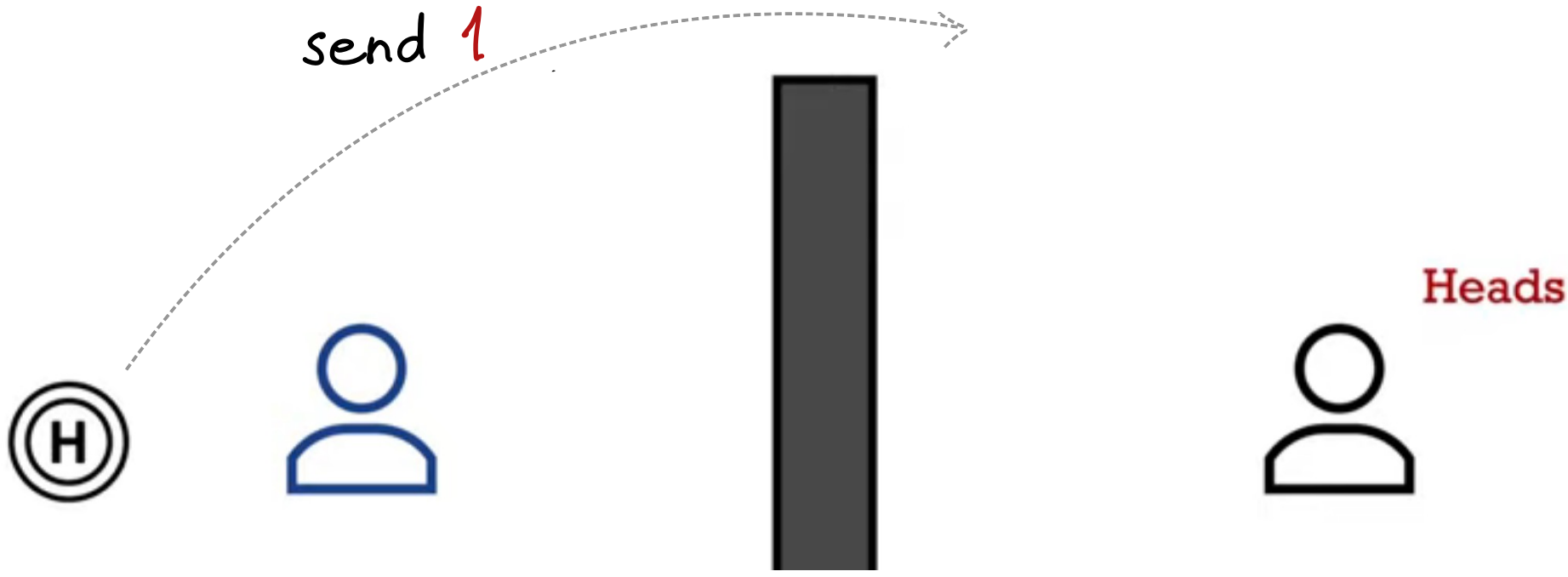
- After observing the outcome of a coin flip, send the relevant bit across the channel to the other person.

P(H) = 0.5

P(T) = 0.5

0 = Tails

1 = Heads

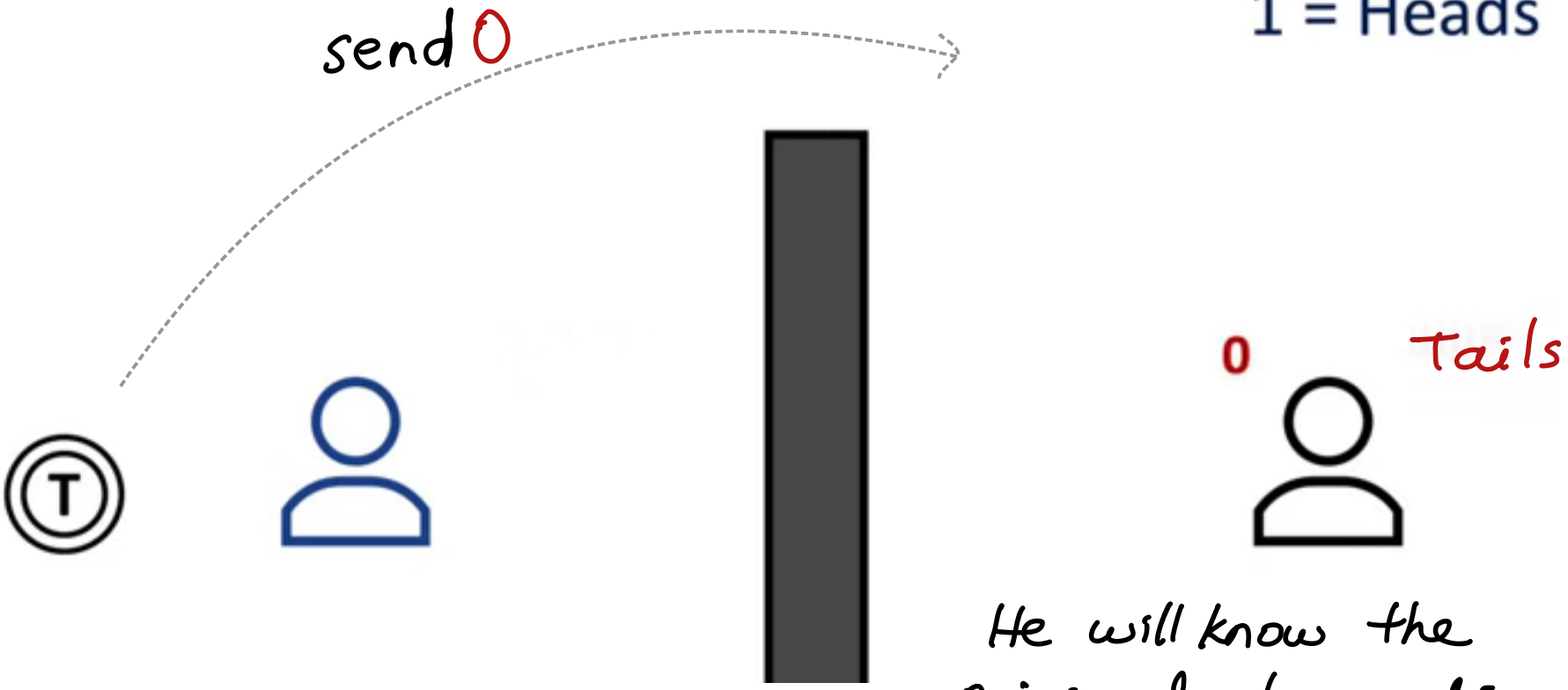


$P(H) = 0.5$

$P(T) = 0.5$

0 = Tails

1 = Heads



- This distribution has an entropy of 1 bit.
- i.e. it takes 1 bit to describe the outcome of an event

P(H) = 0.5

P(T) = 0.5

entropy = 1 bit

0 = Tails

1 = Heads

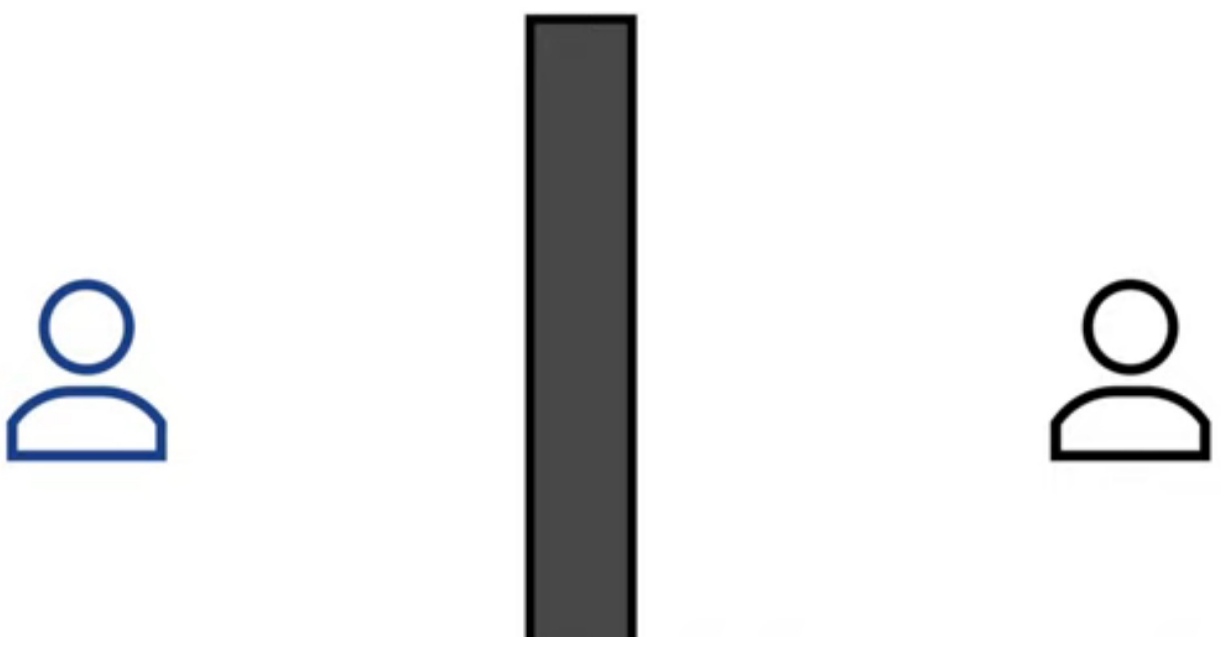


- A random event with 8 possible random outcomes.
- e.g. a tournament with 8 <u>equally strong</u> teams: A, B, C, D, E, F, G, H

P(A) = 0.125	P(E) = 0.125
P(B) = 0.125	P(F) = 0.125
P(C) = 0.125	P(G) = 0.125
P(D) = 0.125	P(H) = 0.125

↓
prob of team D winning

- Prob. of a team winning: $\frac{1}{8}$



- We need 3 bits to describe the outcome
- $2^3 = 8$ different states

P(A) = 0.125	P(E) = 0.125
P(B) = 0.125	P(F) = 0.125
P(C) = 0.125	P(G) = 0.125
P(D) = 0.125	P(H) = 0.125

000 = A
001 = B
010 = C
011 = D
100 = E
101 = F
110 = G
111 = H

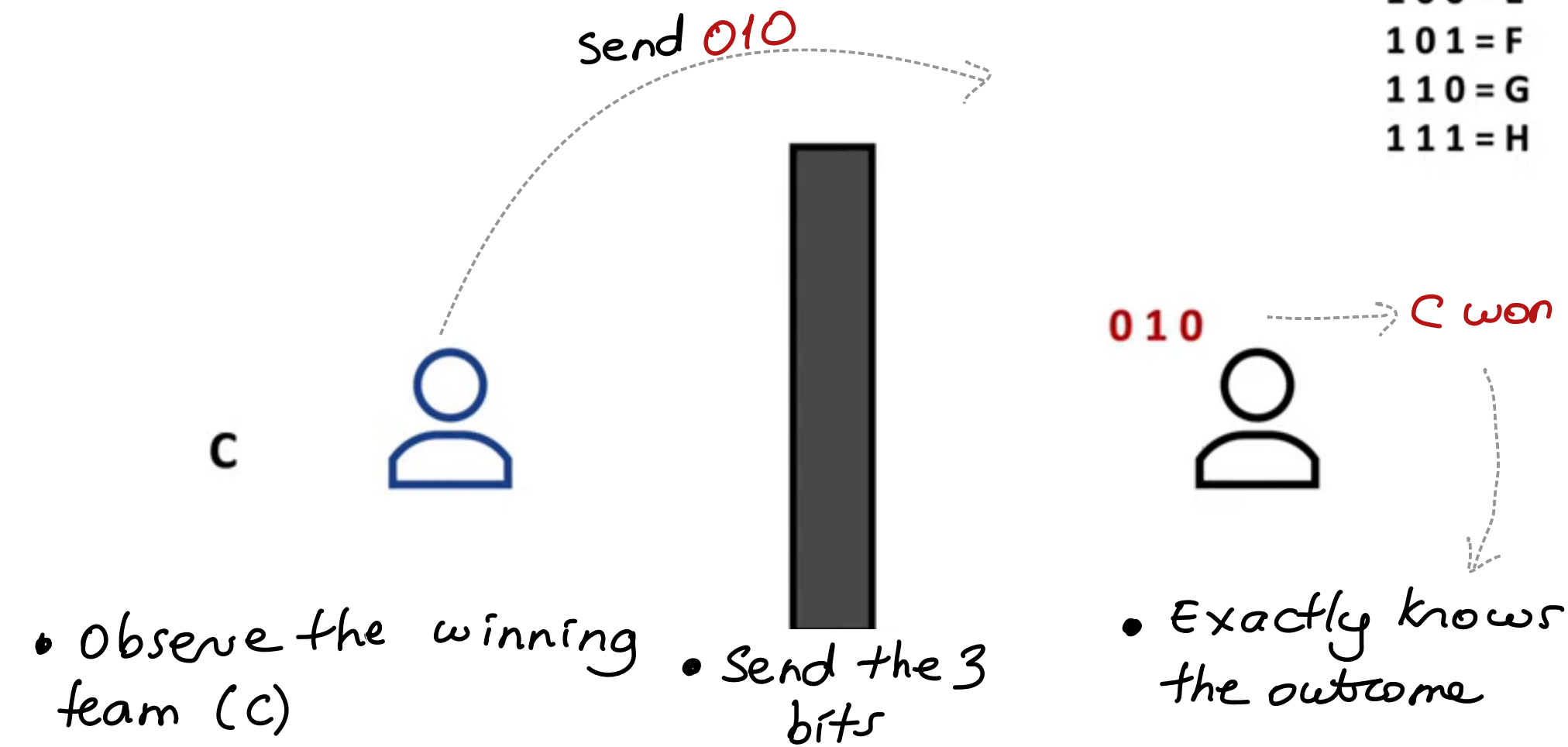






P(A) = 0.125 P(E) = 0.125
P(B) = 0.125 P(F) = 0.125
P(C) = 0.125 P(G) = 0.125
P(D) = 0.125 P(H) = 0.125

000 = A
001 = B
010 = C
011 = D
100 = E
101 = F
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111 = H



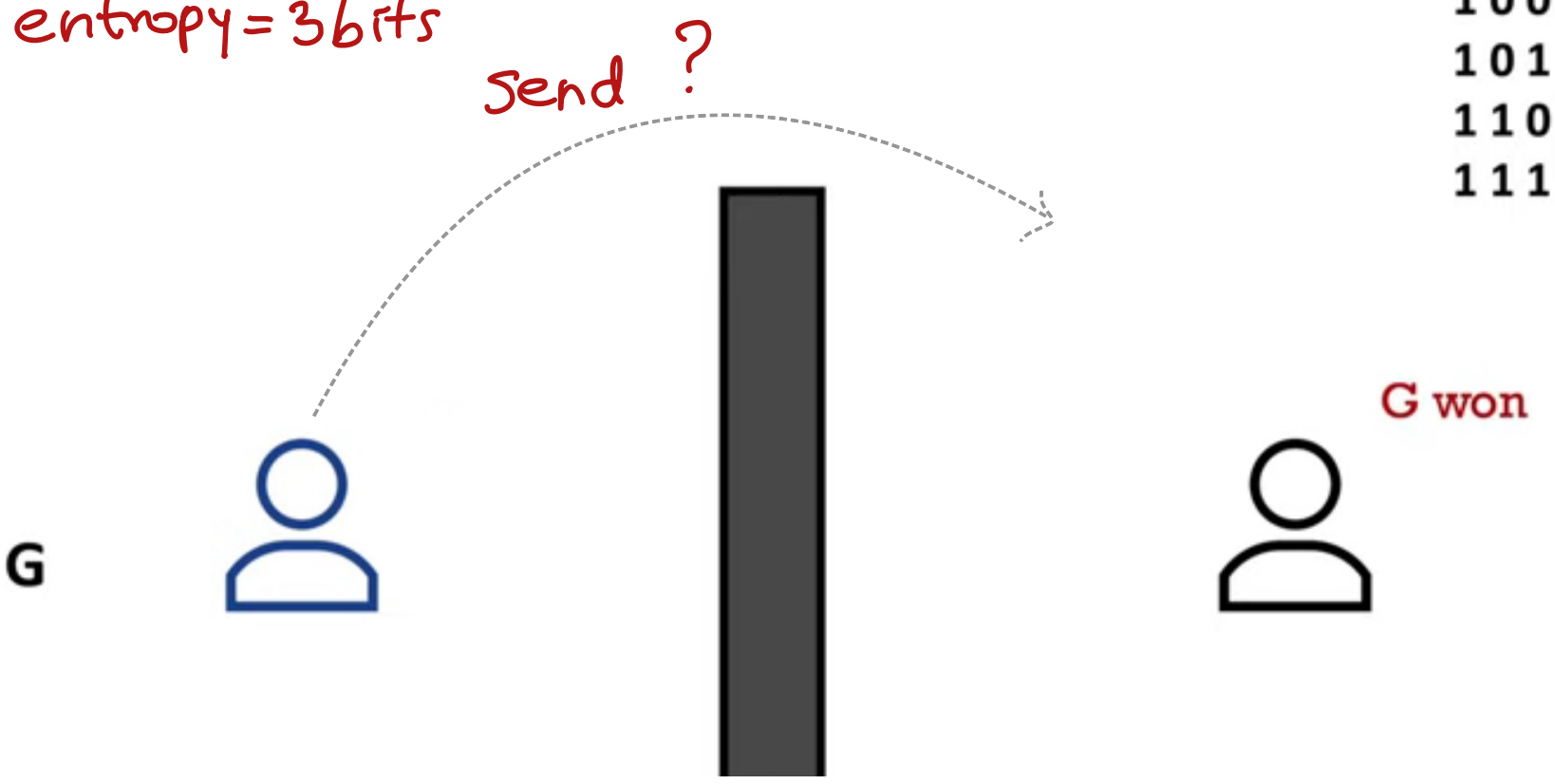
- Observe the winning team (C)
- Send the 3 bits
- Exactly knows the outcome

- This distribution has an entropy of 3 bits.

P(A) = 0.125 P(E) = 0.125
P(B) = 0.125 P(F) = 0.125
P(C) = 0.125 P(G) = 0.125
P(D) = 0.125 P(H) = 0.125

entropy = 3 bits

000 = A
001 = B
010 = C
011 = D
100 = E
101 = F
110 = G
111 = H



G won

Generalize to a uniform distribution with M possible outcomes:

If we have M equally likely outcomes

Then the entropy will be $\log_2 M$ bits

If M is a power of 2, it is trivial to see.

If we have M equally likely outcomes

Then the entropy will be $\log_2 M$ bits

$\log_2 2^N = N$ ⟶ number of bits

↓

M different states

- This formula also holds for uniform distributions where the number of outcomes is not a power of 2.

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

- Let $M = 10$.
- An event with 10 equally likely outcomes

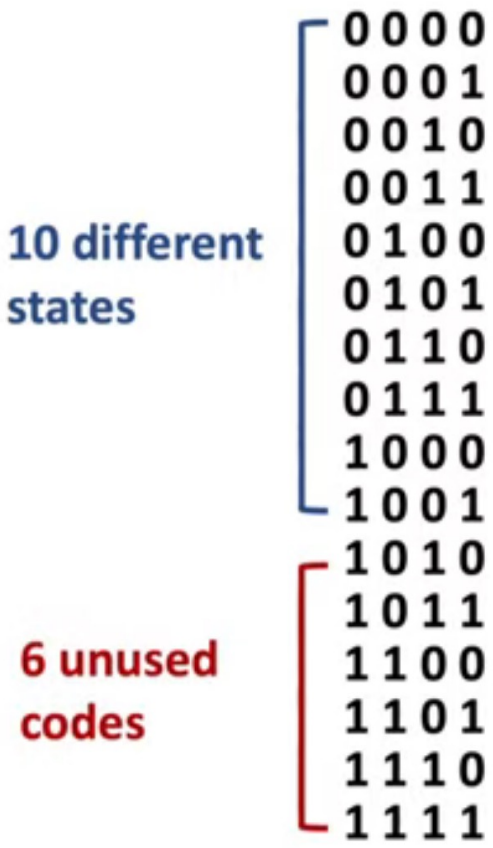
P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

10 different states

0 0 0 0
0 0 0 1
0 0 1 0
0 0 1 1
0 1 0 0
0 1 0 1
0 1 1 0
0 1 1 1
1 0 0 0
1 0 0 1
1 0 1 0
1 0 1 1
1 1 0 0
1 1 0 1
1 1 1 0
1 1 1 1

- 4 bits give 16 unique states, so we could use 4 bits to encode all 10 possible outcomes.

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1



- Inefficient : 6 states are not used.

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

10 different states

0 0 0 0
0 0 0 1
0 0 1 0
0 0 1 1
0 1 0 0
0 1 0 1
0 1 1 0
0 1 1 1
1 0 0 0
1 0 0 1

6 unused codes

1 0 1 0
1 0 1 1
1 1 0 0
1 1 0 1
1 1 1 0
1 1 1 1

A, A, A
A, A, B
A, A, C
...
J, J, H
J, J, I
J, J, J

1000 different triples

- To do better, can encode the outcomes in groups of 3.
- $10 \times 10 \times 10 = 1000$ diffent outcome for groups of 3.

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

10 different states

0000
0001
0010
0011
0100
0101
0110
0111
1000
1001

6 unused codes

1010
1011
1100
1101
1110
1111

0000000000 = A, A, A
0000000001 = A, A, B
0000000010 = A, A, C
...
1111100101 = J, J, H
1111100110 = J, J, I
1111100111 = J, J, J

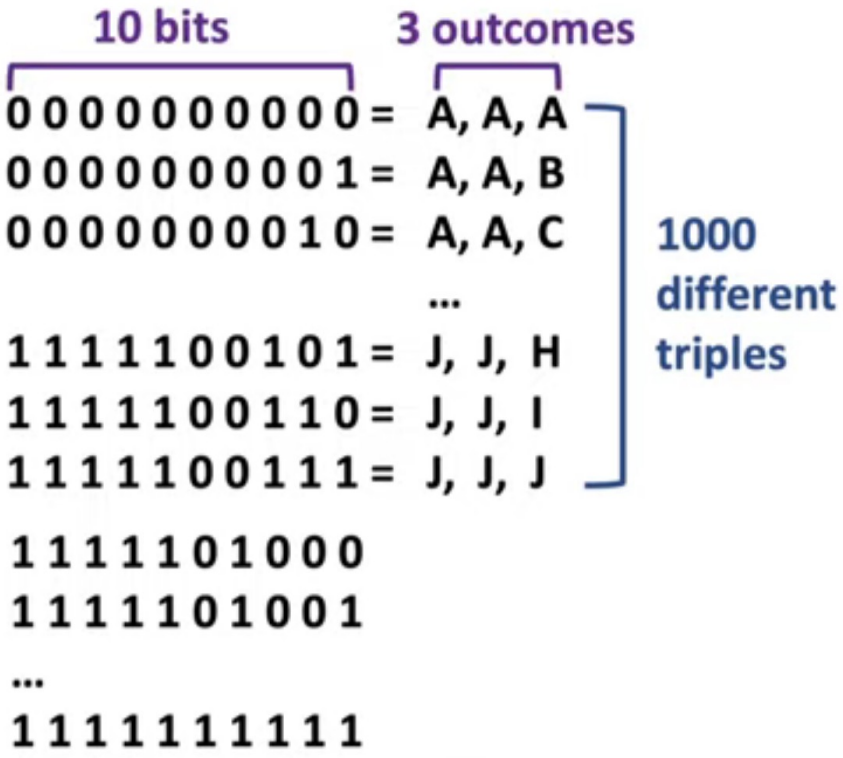
1000 different triples

1111101000
1111101001
...
1111111111

- 10 bits will give us $2^{10} = 1024$ possible states.
- 10 bits are sufficient to encode 1000 different triplets.

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

- we can use 10 bits to encode 3 independent outcomes.

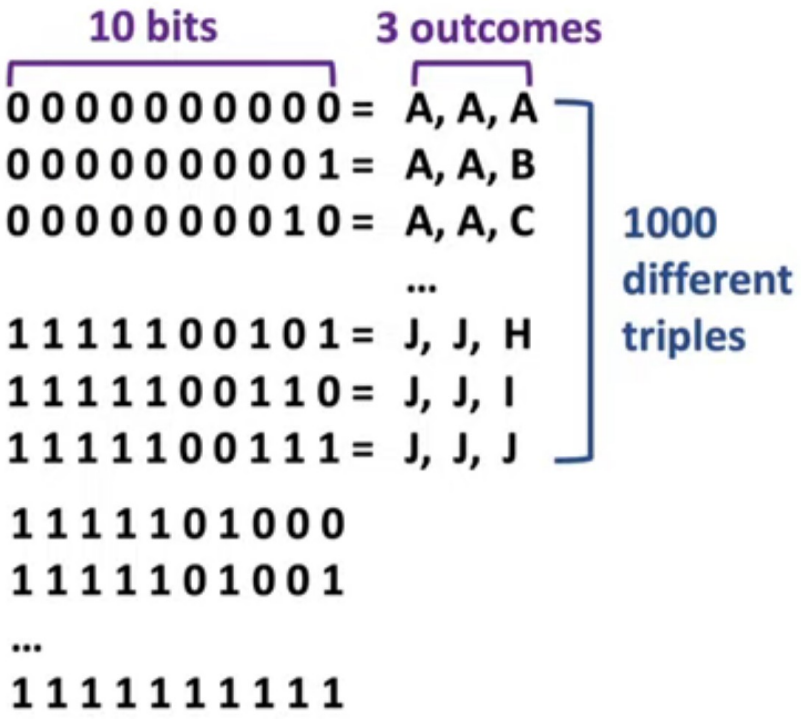


$$\frac{10}{3} = 3.333$$

↓

Average number of bits per outcome.

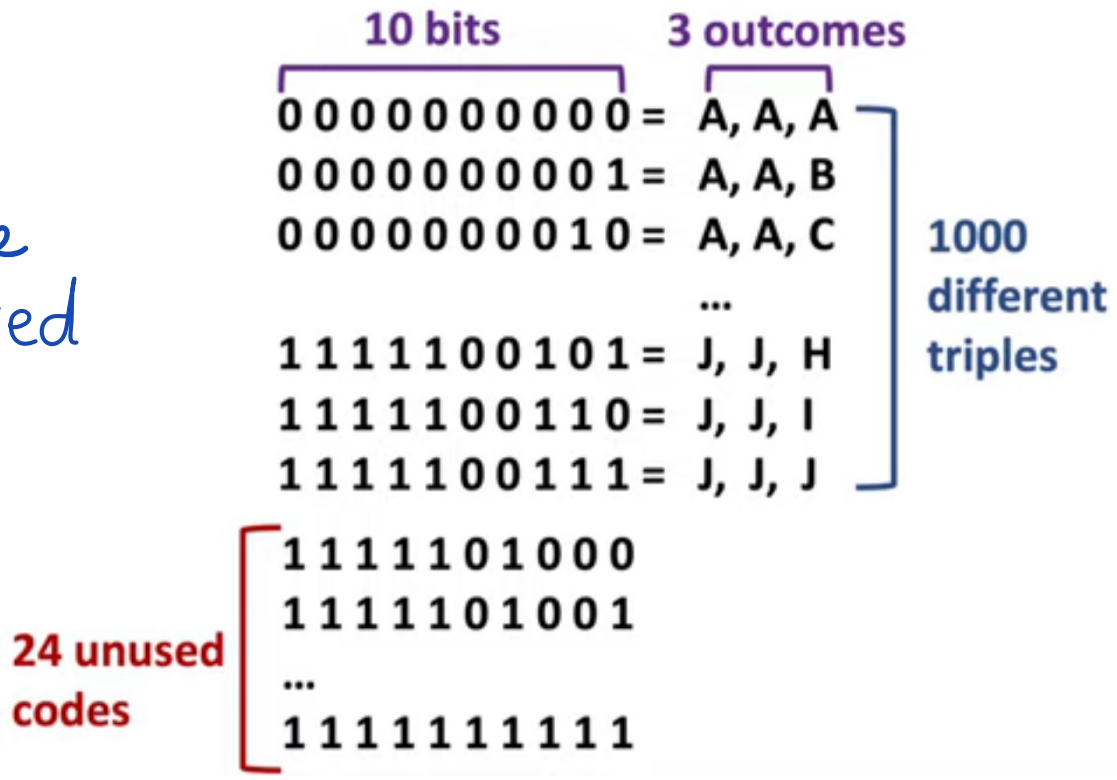
P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1



$$\frac{10}{3} = 3.333$$

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

- Not efficient since we have 24 unused states.

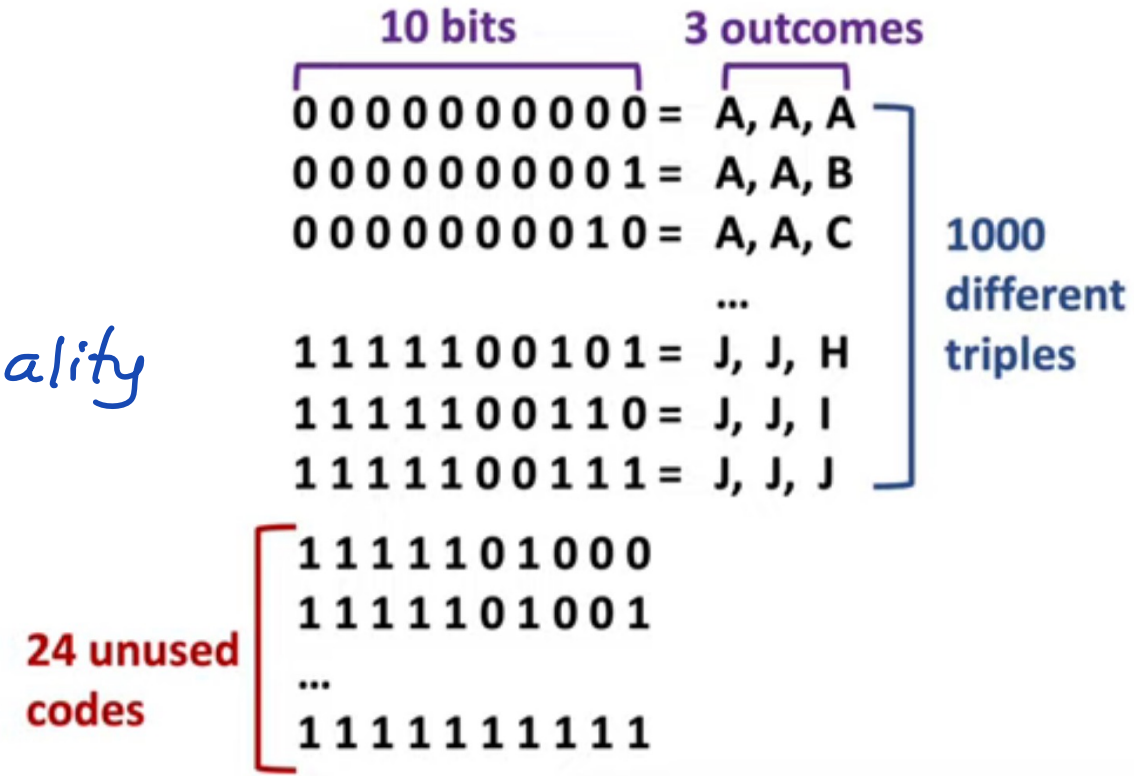


$$\frac{10}{3} = 3.333$$

$$2^B = 10^G$$

- We want this equality the most efficient encoding.
- G: number of grouped observations

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1



$$2^B = 10^G \Rightarrow \log_2 2^B = \log_2 10^G \Rightarrow B = G \log_2 10$$

$$\frac{10}{3} = 3.333$$

$$\frac{B}{G} = \log_2 10$$

Most efficient # of bits to encode each outcome

P(A) = 0.1	P(D) = 0.1	P(G) = 0.1
P(B) = 0.1	P(E) = 0.1	P(H) = 0.1
P(C) = 0.1	P(F) = 0.1	P(I) = 0.1
		P(J) = 0.1

10 bits — 3 outcomes

0 0 0 0 0 0 0 0 0 0 = A, A, A
0 0 0 0 0 0 0 0 0 1 = A, A, B
0 0 0 0 0 0 0 0 1 0 = A, A, C
...
1 1 1 1 1 0 0 1 0 1 = J, J, H
1 1 1 1 1 0 0 1 1 0 = J, J, I
1 1 1 1 1 0 0 1 1 1 = J, J, J

1000 different triples

24 unused codes

1 1 1 1 1 0 1 0 0 0
1 1 1 1 1 0 1 0 0 1
...
1 1 1 1 1 1 1 1 1 1

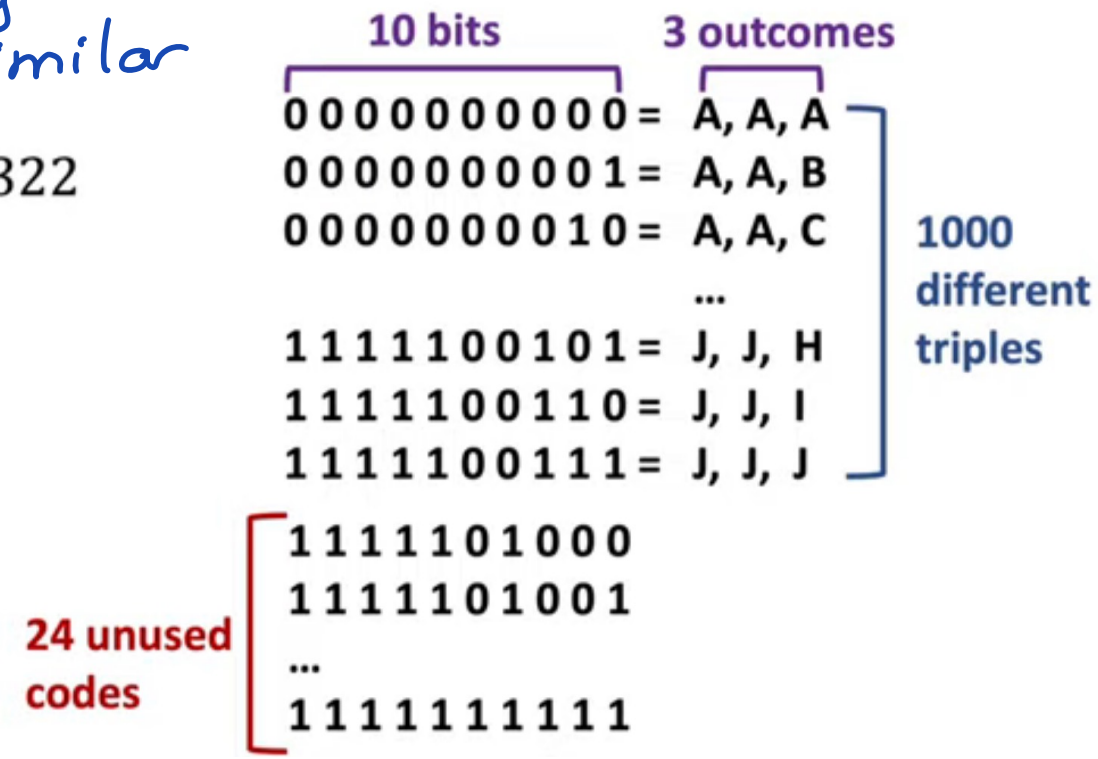
$$\frac{10}{3} = 3.333$$

very similar

$$\frac{B}{G} = \log_2 10 = 3.322$$

Entropy of the uniform distribution.

P(A) = 0.1
P(B) = 0.1
P(C) = 0.1
P(D) = 0.1
P(E) = 0.1
P(F) = 0.1
P(G) = 0.1
P(H) = 0.1
P(I) = 0.1
P(J) = 0.1



$$U(M) \begin{cases} p_1 = \frac{1}{M} \\ p_2 = \frac{1}{M} \\ p_3 = \frac{1}{M} \\ p_4 = \frac{1}{M} \\ \cdots \\ p_M = \frac{1}{M} \end{cases}$$

→ uniform distribution with M outcomes

$$\mathbf{H(U(M)) = \log_2 M}$$

: Entropy of the uniform distribution with M outcomes.

How do we measure the entropy of distributions which are not uniform?

Uniform Distribution:

$$U(M) \begin{cases} p_1 = \frac{1}{M} \\ p_2 = \frac{1}{M} \\ p_3 = \frac{1}{M} \\ p_4 = \frac{1}{M} \\ \cdots \\ p_M = \frac{1}{M} \end{cases}$$

$$\mathbf{H(U(M)) = \log_2 M}$$

Non-uniform distributions:

$$\log_2 \frac{1}{p} \text{ bits} \rightarrow$$

of bits to encode an outcome with probability p.

$$\mathrm{U(M)} \begin{cases} p_1 = \frac{1}{\mathrm{M}} \\ p_2 = \frac{1}{\mathrm{M}} \\ p_3 = \frac{1}{\mathrm{M}} \\ p_4 = \frac{1}{\mathrm{M}} \\ \cdots \\ p_M = \frac{1}{\mathrm{M}} \end{cases}$$

$$\log_2 \frac{1}{p} \text{ bits}$$

simplify

$$= -\log_2 p \text{ bits}$$

$$\mathbf{H\big(U(M)\big) = \log_2 M}$$

$$\mathrm{U(M)} \begin{cases} p_1 = \frac{1}{\mathrm{M}} \\ p_2 = \frac{1}{\mathrm{M}} \\ p_3 = \frac{1}{\mathrm{M}} \\ p_4 = \frac{1}{\mathrm{M}} \\ \cdots \\ p_M = \frac{1}{\mathrm{M}} \end{cases}$$

$$\log_2 \frac{1}{p} \text{ bits}$$

if $p = \frac{1}{M}$

$$= -\log_2 \frac{1}{M} \text{ bits}$$

$$\mathbf{H\left(U(M)\right) = \log_2 M}$$

$$U(M) \begin{cases} p_1 = \frac{1}{M} \\ p_2 = \frac{1}{M} \\ p_3 = \frac{1}{M} \\ p_4 = \frac{1}{M} \\ \cdots \\ p_M = \frac{1}{M} \end{cases}$$

$$\log_2 \frac{1}{p} \text{ bits}$$

$$= \log_2 M \text{ bits} \quad \left(\text{if } p = \frac{1}{M}\right)$$

$$\mathbf{H(U(M)) = \log_2 M}$$

$$U(M) \begin{cases} p_1 = \frac{1}{M} \\ p_2 = \frac{1}{M} \\ p_3 = \frac{1}{M} \\ p_4 = \frac{1}{M} \\ \cdots \\ p_M = \frac{1}{M} \end{cases} \qquad \log_2 \frac{1}{p} \text{ bits}$$

$$= \quad \log_2 M \text{ bits}$$

$$\mathbf{H(U(M)) = \log_2 M}$$

$-\log_2 p_i$: # bits to encode outcome i

- p_i : probability of outcome i for a non-uniform distribution

$$\mathrm{U(M)} \begin{cases} p_1 = \frac{1}{\mathrm{M}} \\ p_2 = \frac{1}{\mathrm{M}} \\ p_3 = \frac{1}{\mathrm{M}} \\ p_4 = \frac{1}{\mathrm{M}} \\ \cdots \\ p_M = \frac{1}{\mathrm{M}} \end{cases}$$

$$\log_2 \frac{1}{p} \text{ bits}$$

$$= \log_2 M \text{ bits}$$

$$\mathbf{H(U(M)) = \log_2 M}$$

$$-\sum_i p_i \log_2 p_i$$

Shannon Entropy: expected # bits over all the outcomes.

Entropy

A *Zero-memory information source* S is a source that emits symbols from an alphabet $\{s_1, s_2, \ldots, s_k\}$ with probabilities $\{p_1, p_2, \ldots, p_k\}$, respectively, where the symbols emitted are statistically independent.

What is the average amount of information in observing the output of the source S?

Call this **Entropy**:

$$H(S) \;=\; \sum_i p_i \cdot I(s_i) \;=\; \sum_i p_i \cdot \log \frac{1}{p_i} \;=\; E_P \left[\log \frac{1}{p(s)} \right]$$

Alternative Explanations of Entropy

$$H(S) \;=\; \sum_i p_i \cdot \log \frac{1}{p_i}$$

1. avg amt of info provided per symbol
2. avg amount of surprise when observing a symbol
3. uncertainty an observer has before seeing the symbol
4. avg # of bits needed to communicate each symbol
   (Shannon: there are codes that will communicate these symbols with efficiency arbitrarily close to $H(S)$ bits/symbol; there are no codes that will do it with efficiency $< H(S)$ bits/symbol)

Entropy as a Function of a Probability Distribution

Since the source S is fully characterized by$P = \{p_1, \ldots p_k\}$ (we don't care what the symbols s_i actually are, or what they stand for), entropy can also be thought of as a property of a probability distribution function P: the avg uncertainty in the distribution. So we may also write:

$$H(S) \;=\; H(P) \;=\; H(p_1, p_2, \ldots, p_k) \;=\; \sum_i p_i \log \frac{1}{p_i}$$

(Can be generalized to continuous distributions.)

Properties of Entropy

$$H(P) \;=\; \sum_i p_i \cdot \log \frac{1}{p_i}$$

1. Non-negative: $H(P) \geq 0$
2. Invariant wrt permutation of its inputs:
   $H(p_1, p_2, \ldots, p_k) = H(p_{\tau(1)}, p_{\tau(2)}, \ldots, p_{\tau(k)})$
3. For any *other* probability distribution $\{q_1, q_2, \ldots, q_k\}$:

$$H(P) \;=\; \sum_i p_i \cdot \log \frac{1}{p_i} \;<\; \sum_i p_i \cdot \log \frac{1}{q_i}$$

4. $H(P) \leq \log k$, with equality iff $p_i = 1/k \;\; \forall i$
5. The further P is from uniform, the lower the entropy.

Special Case: $k = 2$

Flipping a coin with P("head")=p, P("tail")=1-p

$$H(p) = p \cdot \log \frac{1}{p} + (1-p) \cdot \log \frac{1}{1-p}$$

Notice:

- zero uncertainty/information/surprise at edges
- maximum info at 0.5 (1 bit)
- drops off quickly

The Entropy of English

27 characters (A-Z, space).

100,000 words (avg 6.5 characters each)

- Assuming independence between successive characters:
  - uniform character distribution: $\log 27 = 4.75$ bits/character
  - true character distribution: 4.03 bits/character
- Assuming independence between successive *words*:
  - unifrom word distribution: $\log 100,000/6.5 \approx 2.55$ bits/character
  - true word distribution: $9.45/6.5 \approx 1.45$ bits/character
- True Entropy of English is much lower!

Two Sources

Temperature T: a random variable taking on values t

P(T=hot)=0.3

P(T=mild)=0.5

P(T=cold)=0.2

$\Longrightarrow$ H(T)=H(0.3, 0.5, 0.2) = 1.48548

huMidity M: a random variable, taking on values m

P(M=low)=0.6

P(M=high)=0.4

$\Longrightarrow H(M) = H(0.6, 0.4) = 0.970951$

T, M not independent: $P(T = t, M = m) \neq P(T = t) \cdot P(M = m)$

Joint Probability, Joint Entropy

	cold	mild	hot	
low	0.1	0.4	0.1	0.6
high	0.2	0.1	0.1	0.4
	0.3	0.5	0.2	1.0

- $H(T) = H(0.3, 0.5, 0.2) = 1.48548$
- $H(M) = H(0.6, 0.4) = 0.970951$
- $H(T) + H(M) = 2.456431$
- **Joint Entropy**: consider the space of (t, m) events $H(T, M) = \sum_{t,m} P(T = t, M = m) \cdot \log \frac{1}{P(T=t, M=m)}$
  $H(0.1, 0.4, 0.1, 0.2, 0.1, 0.1) = 2.32193$

Notice that $H(T, M) < H(T) + H(M)$!!!

Conditional Probability, Conditional Entropy

$$P(T = t|M = m)$$

	cold	mild	hot	
low	1/6	4/6	1/6	1.0
high	2/4	1/4	1/4	1.0

Conditional Entropy:

- $H(T|M = low) = H(1/6, 4/6, 1/6) = 1.25163$
- $H(T|M = high) = H(2/4, 1/4, 1/4) = 1.5$
- **Average Conditional Entropy** (aka equivocation):
  $H(T/M) = \sum_m P(M = m) \cdot H(T|M = m) =$
  $0.6 \cdot H(T|M = low) + 0.4 \cdot H(T|M = high) = 1.350978$

How much is M telling us on average about T?

$H(T) - H(T|M) = 1.48548 - 1.350978 \approx 0.1345$ bits

Conditional Probability, Conditional Entropy

$$P(M = m|T = t)$$

	cold	mild	hot
low	1/3	4/5	1/2
high	2/3	1/5	1/2
	1.0	1.0	1.0

Conditional Entropy:

- $H(M|T = cold) = H(1/3, 2/3) = 0.918296$
- $H(M|T = mild) = H(4/5, 1/5) = 0.721928$
- $H(M|T = hot) = H(1/2, 1/2) = 1.0$
- Average Conditional Entropy (aka Equivocation):
  $H(M/T) = \sum_t P(T = t) \cdot H(M|T = t) =$
  $0.3 \cdot H(M|T = cold) + 0.5 \cdot H(M|T = mild) + 0.2 \cdot H(M|T = hot) = 0.8364528$

How much is T telling us on average about M?

$H(M) - H(M|T) = 0.970951 - 0.8364528 \approx 0.1345$ bits